

```
avahi-browse _T81._sub._matterc._udp -r
```

A Commissioner can discover Nodes that are currently in Commissioning Mode as a result of a commissioning window opened by a current Administrator as a result of invoking either the [OpenCommissioningWindow](#) command, the [OpenJointCommissioningWindow](#) command or the [OpenBasicCommissioningWindow](#) command, using the presence of the `_CM` subtype as a browsing filter:

```
dns-sd -B _matterc._udp,_CM
```

or

```
avahi-browse _CM._sub._matterc._udp -r
```

4.3.1.15. Efficiency Considerations

Discovering and using an offered service on the network typically involves several steps:

1. Enumeration of instances available on the network ("browsing")
2. Lookup of a selected instance's port number, host name, and other additional information, communicated in DNS-SD using SRV and TXT records ("resolving")
3. Lookup of the IPv6 address(es) associated with the desired target host.
4. Use of IPv6 Neighbor Discovery and/or IPv6 routing to translate from destination IPv6 address to the next-hop link-layer address for that communication.
5. Establishing a cryptographically secure communication channel between the two endpoints, and then engaging in useful communication.

Although the first three steps are exposed in some APIs as separate steps, at a protocol level they usually require only a single network round-trip. When a PTR query is issued to discover service instances, the usual DNS Additional Record mechanism, where packet space permits, automatically places the related SRV, TXT, and address records into the Additional Record section of the reply. These additional records are stored by the client, to enable subsequent steps in the sequence to be performed without additional redundant network operations to learn the same information a second time.

DNS-SD over Multicast DNS works by receiving replies from other Nodes attached to the same local link, Nodes that may have been previously completely unknown to the requester. Because of this, Multicast DNS, like IPv6 Neighbor Discovery, does not have any easy way to distinguish genuine replies from malicious or fraudulent replies. Consequently, application-layer end-to-end security is essential. Should a malicious device on the same local link give deliberately malicious or fraudulent replies, the misbehavior will be detected when the device is unable to establish a cryptographically secure application-layer communication channel. This reduces the threat to a Denial-of-Service attack, which can be remedied by physically removing the offending device.